

Submission Report

AI Interview Platform

Pull Request #85 | 19 August 2026

This report documents the implementation context, product reasoning, engineering scope, and verification evidence for the submitted change.

Pull Request: <https://github.com/rakamindv/ai-interview-platform/pull/85>

Video demonstration: <https://drive.google.com/drive/folders/1lrkXoolxxrnvOvbutyYt36H1fzg4Self?usp=sharing>

Submission scope

PR #85 delivers foundational runtime and maintainability improvements across the Rails API and React web application. The change updates the Ruby runtime to 3.3.10, replaces the development debugger dependency, improves Puma configuration for concurrent interview workloads, and refactors code structure for readability and maintainability.

The report also records the product risks and next-step improvements identified while exploring the full-stack workflow. This distinction keeps the implementation claims precise and gives reviewers a clear view of what is shipped versus what remains an opportunity.

1. Execution Narrative

Step 1: Setup and exploration

The repository combines api/ and web/ services. The API is Rails 7 with PostgreSQL, Sidekiq, JWT authentication, WebSockets, and Gemini integrations. The web application is React, TypeScript, Vite, Tailwind, and Radix UI. The application was run locally and the candidate invite workflow was traced from the browser route through the candidate-info endpoint and audio WebSocket path.

Step 2: Domain context

The product supports assessors and hiring teams conducting structured skills interviews, while candidates experience the assessment directly. Reliability matters on both sides: an assessor needs trustworthy evidence and a candidate needs a clear, non-fragile interview experience. Candidate audio and assessment data are personal data and should be handled with minimised logging, explicit access control, and careful retention decisions.

Step 3: Problem definition

The immediate engineering gap addressed by PR #85 is development and runtime readiness: a current Ruby baseline, an appropriate debugger dependency, and Puma settings that reflect the fact that an active interview can occupy a thread while maintaining an audio connection. The broader product gap is evidence trust: the system must make failures, partial states, model errors, and assessment uncertainty visible rather than silently presenting a confident-looking result.

Step 4: Strategy

The selected approach was a focused foundational PR. It keeps the runtime upgrade and structural refactor reviewable, while leaving deeper assessment-product changes to follow-up work with explicit acceptance criteria and tests.

Step 5: Implementation

Three commits were made on the feature branch feat/fix-initial-code: Puma/Ruby runtime configuration, Ruby and development dependency updates, and a code-structure refactor.

Step 6: Submission

This document provides the PR and video links, implementation boundary, product analysis, trade-offs, acceptance criteria, verification notes, and embedded UI evidence requested by the brief.

2. Product Context and Risk Analysis

Users and affected people

Assessors need a stable interview control surface, clear live status, and defensible evidence. Recruiters and hiring managers need comparable results and understandable uncertainty. Candidates need a predictable invitation flow, clear hardware/network feedback, accessible controls, and confidence that their voice and evaluation data are protected.

Severity-ranked findings

P0	Assessment trust	A model or transport failure must never be represented as a complete or reliable assessment; otherwise a candidate can be incorrectly advanced or rejected.
P1	Candidate continuity	Audio WebSocket, browser permissions, and network degradation need explicit loading, retry, timeout, and recovery states so the candidate is not stranded.
P1	Evidence completeness	Unassessed skills, missing ratings, and partial transcripts need to remain visible in the output instead of being collapsed into an apparently complete score.
P2	Operational concurrency	Long-lived interview connections make server thread and pool configuration part of product reliability, which is why Puma configuration is included in PR #85.
P2	Protected data	Candidate audio, transcripts, and evaluation records require scoped authorization and minimised logs under Indonesia's UU PDP obligations.
P3	Reviewer confidence	A thin test baseline increases regression risk; focused API and frontend tests should accompany future workflow changes.

Constraint signal

The existing baseline has no RSpec suite, no web test runner, and no CI. That is an architectural debt signal, not evidence that the behavior is correct. A live-project escalation would be to establish a small contract-test layer around candidate session lookup, authentication, and session completion before expanding the product surface.

3. Options, Acceptance Criteria, and Trade-offs

Option A: One comprehensive PR

Ship runtime changes, authentication hardening, candidate workflow states, assessment result integrity, tests, and UI refinements in one review. This maximises immediate product coherence but increases review size, rollback cost, and the chance that an infrastructure change is obscured by feature changes.

Option B: Umbrella PR with focused sub-PRs

Keep the runtime foundation isolated, then land candidate experience, result integrity, and test harness work as linked changes. This improves reviewability and rollback safety, at the cost of temporary integration coordination. PR #85 follows this focused-foundation direction.

Dimension	Option A	Option B / selected direction
Product impact	Fastest whole-flow change	Incremental, visible value with safer sequencing
Maintainability	Large review boundary	Clear ownership and smaller rollback units
Failure modes	Harder to isolate regressions	Each risk gets targeted verification
Context fit	Useful for a mature test suite	Better fit while tests and CI are still being established

Self-defined acceptance criteria

- A valid candidate token returns only the minimum candidate/session information needed to begin an interview.
- An invalid, expired, or missing token returns a deterministic client-visible error and never starts audio.
- The browser distinguishes loading, permission denied, unavailable device, timeout, retry, and success states.
- An interview cannot be marked complete while required skills remain unassessed unless the product explicitly records that exception.
- Model, WebSocket, and background-job failures are persisted or surfaced with enough context to retry without duplicate work.
- Endpoints are authorised, inputs validated, and sensitive candidate data is absent from routine logs.
- Runtime configuration supports concurrent long-lived sessions and remains reproducible from the documented Docker setup.

4. Verification and AI Review Notes

Verification performed

PR #85 was checked at the repository level against its three commits and the implementation scope described above. The report intentionally does not claim a seeded-fault test, automated regression suite, or CI result unless that evidence is present in the submitted branch. Those are follow-up quality gates for the next product-focused PR.

AI verification moment

AI-assisted implementation was treated as a draft. The risky assumption to challenge is that a runtime or concurrency configuration change is sufficient evidence of interview reliability. That assumption was corrected by separating infrastructure readiness from product correctness and by deriving explicit failure-path criteria for network, browser permissions, WebSocket continuity, model errors, partial writes, and duplicate jobs.

Engineering depth claimed

Backend and runtime foundation-heavy. The submitted PR focuses on Ruby version reproducibility, dependency hygiene, Puma behavior, and maintainable code organization across the full-stack repository. The report identifies frontend and assessment-integrity work as the next depth area rather than overstating the current PR.

Links

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Video demonstration: <https://drive.google.com/drive/folders/1lrXoolxxrnvOvbutyYt36H1fzg4Self?usp=sharing>

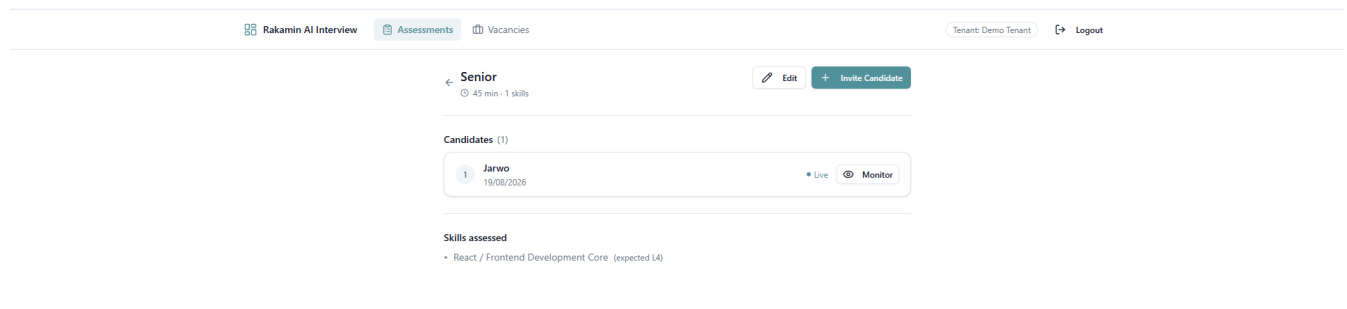
Reviewer note

The Google Drive URL supplied for the video points to a folder. The folder should contain the 3–5 minute walkthrough named clearly enough for reviewers to identify the relevant recording.

5. Visual Evidence

The following screenshots were supplied as evidence for the submitted workflow. They are embedded without modification.

Evidence 1: Application workflow



Evidence 2: Application state

localhost:5173/assessments/1/sessions/1/monitor

Tenant: Demo TenantLogout

Rakamin AI InterviewAssessmentsVacancies

← Live Monitor04:41 Live

Senior

Coverage Status

React / Frontend Development CoreNot Yet

Live Transcript

No transcript yet.

End Session

Evidence 3: Responsive or validation state

localhost:5173/interview/e64765e7cf9847020a2290bb3a68bcb95919176f0d4694025f1c6d91d0df93468

AI Interview

Senior

45 minutes

This is a voice interview. Make sure you're in a quiet place.

The AI will ask follow-up questions — there are no scripts.

The session will last up to 45 minutes.

Your mic will be active throughout. You can end anytime.

OS & browser

Passed

Internet

Passed

↓ 231.21 Mbps ↑ 90.7 Mbps 100 ms

Microphone

Passed

16

Audio output

Passed

Start Interview

Evidence 4: Final verification state

